

Non-mandatory exercises: C

More types

```
1. """
Icelandic_literature
"""

name = "Táten_om_Guld-Ásas_Pórð"
intro_text = "På_den_tid_då_kung_Eystein,_son_till_
    Magnus_Barfot,_regerade_kom_det_från_Island_en_man_
    som_hette_Pórð._Han_härstammade_från_Östfjordarna_
    och_var_fattig._Han_var_en_lovande_man_och_långt_
    bättre_än_sin_far_och_stod_sig_väl_i_jämförelse_med_
    sina_likar._Han_var_en_god_skald._När_han_kom_till_
    staden_hade_han_inga_pengar_till_husrum._På_
    kvällen_kom_han_till_ett_hus_som_ägdes_av_en_kvinna_
    som_hette_Ása..."

nr_of_words = 257
mean_word_len = 4.5
verse = False
author = None
translated = True
translator = "Sofia_Tingsell"

print(type(name))

print(type(mean_word_len))

print(type(verse))

print(type(author))

print(type(translated))
```

The first `print` will result in `<class 'str'>` being printed.
What will the rest of the code print?

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2.

```
print(2 > 1)

print(2 < 1)

print(2 >= 1)

print(2 <= 1)

print(1 < 2)

# i.e. does "a" come before "b" in the alphabet?
print("a" < "b")

print(2 <= 2)
```

The first line above will result in that the boolean `True` is printed out. What print-outs will the rest of the code result in?

3.

```
print(type(2 > 1))
print(type(2 < 1))
print(type(2 >= 1))
print(type(2 <= 1))
print(type(1 < 2))

# i.e. does "a" come before "b" in the alphabet?
print(type("a" < "b"))

print(type(2 <= 2))
```

All prints will print out the same thing. What do they print?

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1 if-else-statements and relational operators

4. What is printed?

```
time = 20

if time < 12:
    print("Good_morning!")
else:
    print("Hello!")
```

5. We can of course use this in a function.

```
def print_greeting(time):
    if time < 12:
        print("Good_morning!")
    else:
        print("Hello!")

print_greeting(10)
print_greeting(12)
print_greeting(18)
```

What is printed?

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6. Equals is written as `==`. What is printed?

```
def print_greeting(time):  
    if time == 12:  
        print("Time_for_lunch!")  
    else:  
        print("Hello")  
  
print_greeting(10)  
print_greeting(12)  
print_greeting(14)
```

7. Not equals is written as `!=`. What is printed?

```
def print_greeting(time):  
    if time != 12:  
        print("Hello")  
    else:  
        print("Time_for_lunch!")  
  
print_greeting(10)  
print_greeting(12)  
print_greeting(14)
```

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8. Equals also works for strings. What is printed?

```
def print_good_morning(name):  
    if name == "Araki":  
        print("Ohayou_gozaimasu_" + name + "_sensei")  
    else:  
        print("Ohayou_" + name + "_san")  
  
print_good_morning("Kitajima")  
  
print_good_morning("Araki")  
  
print_good_morning("Jonas")
```

9. Not equals also works for strings. What is printed?

```
def print_good_morning(name):  
    if name != "Araki":  
        print("Ohayou_" + name + "_san")  
    else:  
        print("Ohayou_gozaimasu_" + name + "_sensei")  
  
print_good_morning("Kitajima")  
  
print_good_morning("Araki")  
  
print_good_morning("Jonas")
```

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10. You can also check the value of a boolean. What is printed?

```
def print_greeting(is_hungry):  
    if is_hungry:  
        print("Hi! Let's go for lunch!")  
    else:  
        print("Hello! How are you?")  
  
print_greeting(True)  
print_greeting(False)
```

11. If you don't need an else condition, just leave it out. What is printed here?

```
def print_greeting(is_hungry):  
    if is_hungry:  
        print("Hi! Let's go for lunch!")  
  
print_greeting(True)  
print_greeting(False)
```

12. What is printed here?

```
def print_good_morning(name):  
    if name != "Araki":  
        print("Ohayou" + name + "san")  
  
print_good_morning("Kitajima")  
  
print_good_morning("Araki")  
  
print_good_morning("Jonas")
```

13. Write a function with your own greeting message in some language. Vary what is printed depending on the argument(s) passed to the function.

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2 if-else-statements and logic operators

14. What is printed here?

```
def print_greeting(is_hungry, time):  
    if is_hungry and time == 12:  
        print("Hi! Let's go for lunch!")  
    else:  
        print("Hello!")  
  
print_greeting(True, 12)  
  
print_greeting(False, 12)  
  
print_greeting(True, 11)  
  
print_greeting(False, 11)
```

15. What is printed here?

```
def print_greeting(is_hungry, time):  
    if is_hungry or time == 12:  
        print("Hi! Let's go for lunch!")  
    else:  
        print("Hello!")  
  
print_greeting(True, 12)  
  
print_greeting(False, 12)  
  
print_greeting(True, 11)  
  
print_greeting(False, 11)
```

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16. What is printed?

```
def print_greeting(is_hungry, time):
    if is_hungry and time >= 11 and time <= 13:
        print("Hi!_Let's_go_for_lunch!")
    else:
        print("Hello!")

print_greeting(True, 12)
print_greeting(False, 12)
print_greeting(True, 11)
print_greeting(False, 11)
print_greeting(True, 13)
```

17. What is printed?

```
def print_greeting(t):
    if (t > 11 and t < 13) or (t > 17 and t < 19):
        print("Hi!_Let's_have_something_to_eat!")
    else:
        print("Hello!")

time = 12
print_greeting(time)
time = time + 2
print_greeting(time)
time = time + 4
print_greeting(time)
```


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18. What is printed here?

```
def print_good_morning(name):
    if name == "Araki" or name == "Rzepka":
        print("Ohayou_gozaimasu_" + name + "_sensei")
    else:
        print("Ohayou_" + name + "_san")

print_good_morning("Kitajima")

print_good_morning("Araki")

print_good_morning("Jonas")

print_good_morning("Rzepka")
```

19. What is printed?

```
def print_good_morning(name):
    if name != "Araki" and name != "Rzepka":
        print("Ohayou_" + name + "_san")
    else:
        print("Ohayou_gozaimasu_" + name + "_sensei")

print_good_morning("Kitajima")

print_good_morning("Araki")

print_good_morning("Jonas")

print_good_morning("Rzepka")
```

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20. What is printed?

```
def print_greeting_hangy(is_hungry):  
    if not(is_hungry):  
        print("Hello!_What_a_fine_day!")  
        # Grumpy when hungry, so no greeting  
  
print_greeting_hangy(True)  
print_greeting_hangy(False)
```

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3 Functions that return booleans

21. What is printed?

```
# Not typical Python code, but possible
def is_morning(time):
    if time < 12:
        return True
    else:
        return False

if is_morning(10):
    print("Good_morning")
if is_morning(14):
    print("Good_morning")
if not(is_morning(16)):
    print("Hello!")
```

22. What is printed?

```
# Also not typical Python code, but possible
def is_morning(time):
    is_time_early = time < 12
    return is_time_early

if is_morning(10):
    print("Good_morning")
if is_morning(14):
    print("Good_morning")
if not(is_morning(16)):
    print("Hello!")
```

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23. What is printed?

```
# More typical Python code
def is_morning(time):
    return time < 12

if is_morning(10):
    print("Good_morning")
if is_morning(14):
    print("Good_morning")
if not(is_morning(16)):
    print("Hello!")
```

24. What is printed?

```
def is_time_for_lunch(t):
    return t >= 11 and t <= 13

time = 10
if is_time_for_lunch(time):
    print("Time_to_eat")
time = time + 2
if is_time_for_lunch(time):
    print("Time_to_eat")
```

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25. What is printed?

```
def is_time_for_lunch(t, has_eaten):  
    return t >= 11 and t <= 13 and not(has_eaten)  
  
time = 10  
has_eaten = False  
  
if is_time_for_lunch(time, has_eaten):  
    print("Time to eat")  
  
time = time + 1  
if is_time_for_lunch(time, has_eaten):  
    print("Time to eat")  
  
time = time + 1  
has_eaten = True  
if is_time_for_lunch(time, has_eaten):  
    print("Time to eat")
```

26. What is printed?

```
def is_morning(time):  
    return time < 12  
  
print(is_morning(10))  
print(is_morning(11))  
print(is_morning(14))  
  
is_currently_morning = is_morning(16)  
print(is_currently_morning)
```

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Modulo

Returns the remainder of a division

27. What is printed for the last two prints?

```
print(2%2) # --> 0
print(3%2) # --> 1
print(4%2) # --> 0
print(5%2) # --> 1
print(6%2)
print(7%2)
```

28. What is printed?

How could the structure of the code below be improved?

```
nr = 4
if nr % 2 == 0:
    print(str(nr) + "_is_even")

nr = nr + 1
if nr % 2 == 0:
    print(str(nr) + "_is_even")

nr = nr + 1
if nr % 2 == 0:
    print(str(nr) + "_is_even")
```

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29. What is printed?

```
print(10 % 10 == 0)
print(15 % 10 == 0)
print(20 % 10 == 0)
print(25 % 10 == 0)
print(100 % 10 == 0)
```